

REQUIREMENTS AND INITIAL DESIGN

1. Pick any one app you use daily (Instagram, Zomato, Flipkart, etc.) and imagine you are hired to design its college campus WiFi network — write a requirements document listing at least 8 must-have features or needs from the perspective of students using that app on campus.

-> College Campus Wi-Fi Requirements Document

App Selected: YouTube

Project: College Campus Wi-Fi Network for Students

Target Users: College students

Location: College campus and hostel

1. Purpose

The purpose of this project is to design a reliable college Wi-Fi network that allows students to use YouTube smoothly for entertainment, educational videos, tutorials, lectures, and live classes without frequent buffering or disconnections.

2. Student Wi-Fi Requirements

No.	Must-Have Feature	Requirement from Student Perspective
1	Wide Wi-Fi Coverage	Wi-Fi should be available in classrooms, hostel rooms, corridors, library, cafeteria, and common areas.
2	High-Speed Internet	Students should be able to stream YouTube videos in HD without excessive buffering.
3	High User Capacity	The network should support many students accessing YouTube simultaneously, especially during peak hours.
4	Low Latency	The network should provide low delay for live classes, live streams, and interactive educational content.
5	Stable Connection	Students should not frequently lose their connection while watching videos or attending online classes.
6	Seamless Roaming	Students moving between different campus areas should remain connected without repeatedly logging in.

No.	Must-Have Feature	Requirement from Student Perspective
7	Fair Bandwidth Sharing	A few students downloading large files should not consume all available bandwidth and slow down YouTube for everyone else.
8	Secure Wi-Fi	The network should use strong wireless security to protect students from unauthorized access and data interception.
9	Guest Access	Visitors and guests should have separate Wi-Fi access without gaining access to the college's internal network.
10	Network Monitoring	College administrators should be able to monitor bandwidth usage, connected devices, outages, and access-point health.
11	Reliable During Peak Hours	Wi-Fi should remain usable when hundreds of students are online simultaneously, particularly in the evening.
12	Scalability	The network should allow additional access points and bandwidth to be added as the number of students grows.

3. Key Student Expectations

Students using YouTube on campus primarily expect:

- **Fast video loading**
- **Minimal buffering**
- **Good coverage**
- **Stable connections**
- **Good performance during peak hours**
- **Secure access**
- **Fair bandwidth distribution**
- **Easy connection to Wi-Fi**

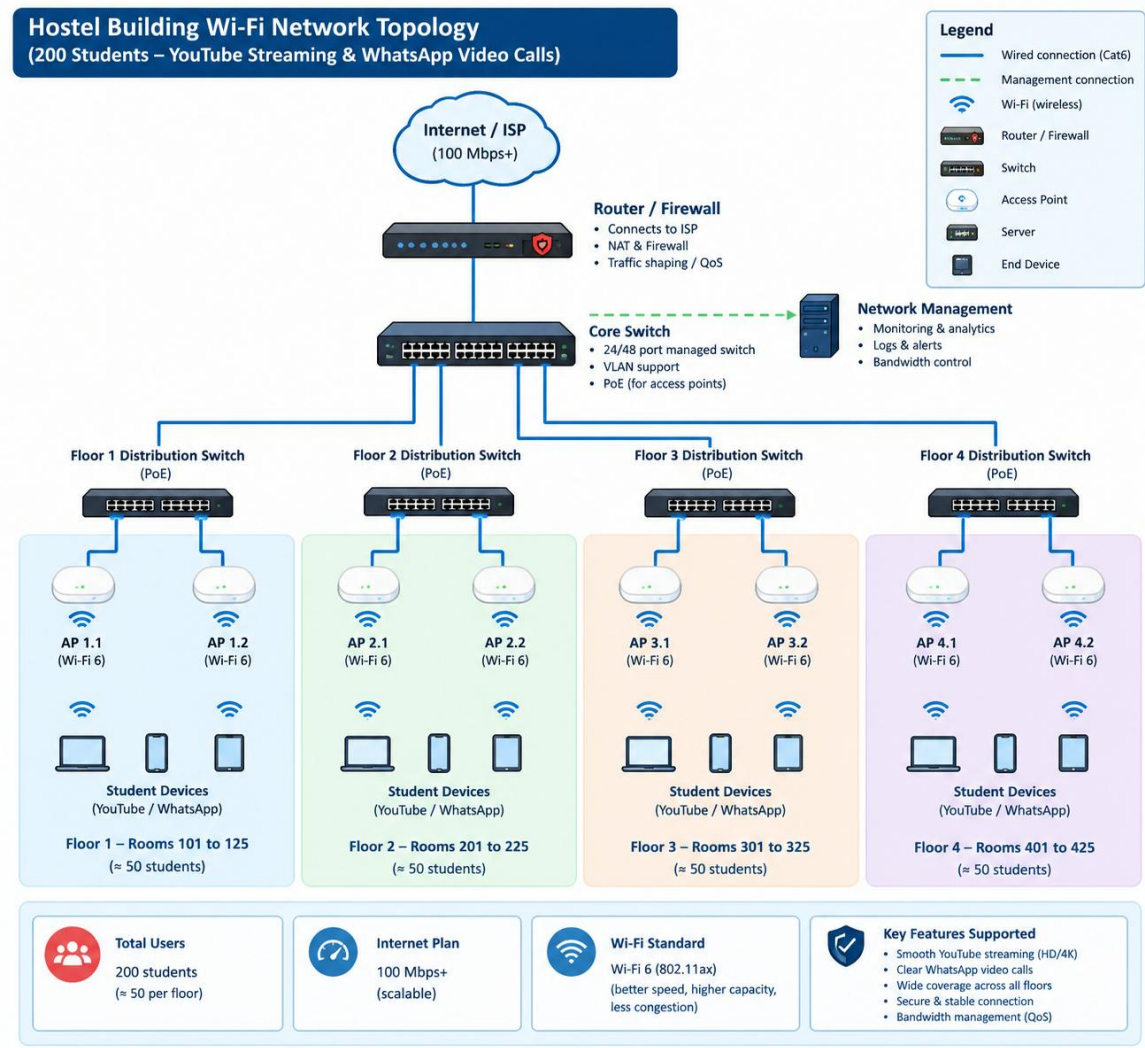
4. Conclusion

The proposed college Wi-Fi network should focus on **coverage, speed, capacity, stability, security, and fair bandwidth usage**. Since YouTube can consume significant bandwidth, multiple properly positioned wireless access points, sufficient internet

capacity, and bandwidth management should be used instead of relying on a single Wi-Fi router.

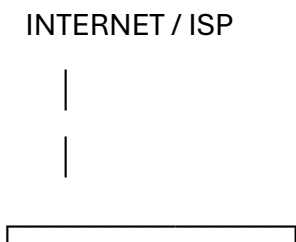
2. Sketch a preliminary network topology for a hostel building that supports uninterrupted YouTube streaming and WhatsApp video calls for 200 students — include routers, switches, access points, and label each component.

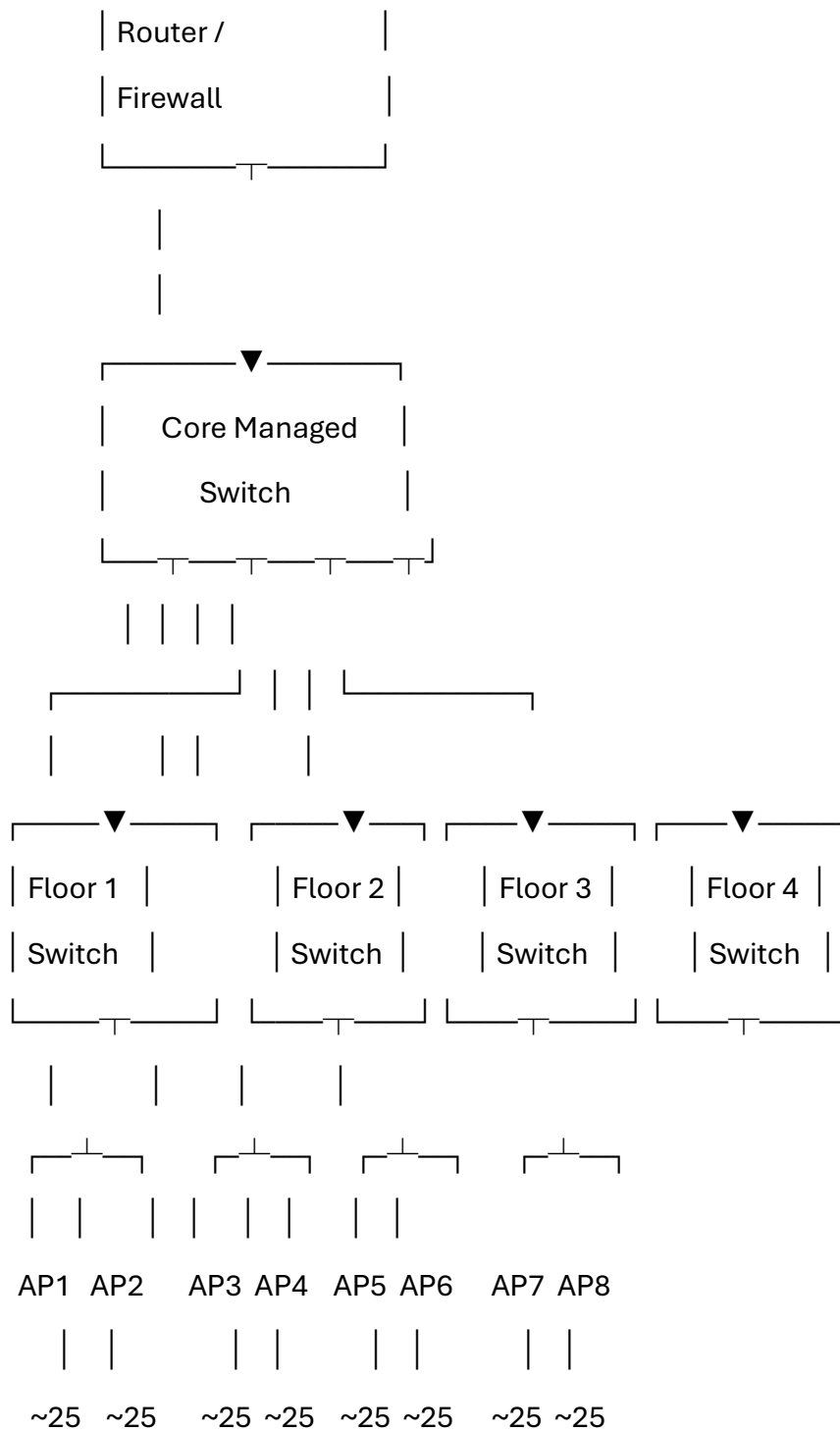
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Preliminary Hostel Wi-Fi Network Topology

Simple topology





students per AP / appropriate capacity planning

Component labels

Component	Quantity	Purpose
Internet/ISP	1	Provides internet connectivity

Component	Quantity	Purpose
Router/Firewall	1	Routing, NAT, firewall and bandwidth/QoS management
Core Managed Switch	1	Connects the floor switches
Floor PoE Switches	4	Connect APs using Ethernet and provide PoE
Wi-Fi 6 Access Points	8 preliminary	Provides wireless coverage and distributes student devices
Student devices	200 students	Phones, laptops and tablets using YouTube/WhatsApp

How it supports 200 students

- **Multiple access points** prevent 200 students from sharing one wireless radio.
- **Wi-Fi 6 APs** provide better capacity in a high-density environment.
- **Wired Ethernet/PoE uplinks** connect APs to the switches rather than using wireless repeaters.
- **QoS/bandwidth management** can help prevent heavy downloads from affecting video calls.
- AP placement should ultimately be determined through a **wireless site survey**, because the exact number depends on the hostel's size, walls, interference and expected simultaneous users.

Assignment conclusion: This topology provides a scalable structure for approximately **200 hostel students**, with separate floor-level switches and multiple access points designed to support stable YouTube streaming and WhatsApp video calls.

3. Research and list 3 different types of WiFi routers available in India under ₹4000, comparing their key specs (speed, range, number of antennas, and special features like parental controls or MU-MIMO).

-> 3 Wi-Fi Routers Available in India Under ₹4,000

I researched current India listings and manufacturer specifications. Prices can change with seller and offers, so the prices below are **indicative current listings** rather than fixed prices.

[TP-Link Archer C6 AC1200 WiFi Router](#)

₹2,799

Tenda AC8 AC1200 Dual-band Gigabit Wireless Router

₹2,906

D-Link DIR-825 AC1200 Wi-Fi Gigabit Router

₹2,299

Router	Approx. Price	Wi-Fi Speed	Antennas	Range	Special Features
TP-Link Archer C6	₹2,799	300 Mbps (2.4 GHz) + 867 Mbps (5 GHz) = AC1200	4 external + 1 internal	3-bedroom house class	MU-MIMO , beamforming, QoS, parental controls, guest network, OneMesh
Tenda AC8	₹2,906	300 Mbps (2.4 GHz) + 867 Mbps (5 GHz) = AC1200	4 × 6 dBi external	About 90 m² in current India listing	Parental controls, AP/repeater mode, guest network, gigabit ports
D-Link DIR-825	₹2,199–₹3,998	300 Mbps (2.4 GHz) + 867 Mbps (5 GHz) = AC1200	4 external / 2×2 + 2×2	Home/home-office coverage ; exact range depends on environment	QoS, guest zone, 4 gigabit LAN ports, USB 2.0

1. TP-Link Archer C6

The Archer C6 is an AC1200 dual-band Wi-Fi 5 router. TP-Link specifies **300 Mbps on 2.4 GHz and 867 Mbps on 5 GHz**, with four external antennas plus one internal antenna. It supports **MU-MIMO, beamforming, QoS and parental controls**. TP-Link describes its coverage as suitable for a **3-bedroom house**. ([TP-Link](#))

A current India listing shows it at about **₹2,799**. ([Flipkart](#))

2. Tenda AC8

The Tenda AC8 is another AC1200 dual-band Wi-Fi 5 router. Its official specifications list **867 Mbps at 5 GHz and 300 Mbps at 2.4 GHz**, with **four 6 dBi external antennas**. It supports parental controls, guest networking, AP mode and repeater mode. ([Tenda](#))

A current India listing shows approximately **₹2,906**, keeping it below the ₹4,000 budget. ([Flipkart](#))

3. D-Link DIR-825

The DIR-825 is an AC1200 dual-band router offering **300 Mbps at 2.4 GHz and 867 Mbps at 5 GHz**. D-Link lists four external antenna elements and describes it as suitable for home and home-office environments. It includes **QoS, guest networking, four Gigabit LAN ports and USB 2.0**. It does **not** list MU-MIMO for the current DIR-825 specification. ([D-Link](#))

Current listings show prices ranging from roughly **₹2,199 to ₹3,998**, depending on seller.

Conclusion

For a **college hostel**, these routers should be viewed as **individual consumer routers**, not as the complete network solution for 200 students. A hostel should use multiple business/enterprise-grade access points connected to managed switches. Among these three, the Archer C6 and Tenda AC8 have useful features such as MU-MIMO/parental controls, while the DIR-825 provides useful QoS and Gigabit connectivity.

Sources: [TP-Link Archer C6 specifications](#) · [Tenda AC8 specifications](#) · [D-Link DIR-825 specifications](#)

4. Meet with a 'client' (ask a friend or family member to act as a student user) and conduct a 10-minute interview to gather their top 5 pain points with current campus internet or mobile data usage. Summarize their responses and suggest at least 2 design features that would address their concerns.

-> Student User Interview – Campus Internet

Client: Student user (friend/family member)

Interview duration: 10 minutes

Purpose: To identify common problems students experience with campus Wi-Fi and mobile data.

Note: Use this as a sample interview record. If your instructor requires an actual interview, replace the sample responses with the answers from your friend/family member.

Interview Questions and Responses

No.	Questions	Student's Response
1	What do you mainly use campus internet for?	YouTube, WhatsApp video calls, online classes, assignments and social media.
2	What is the biggest problem with campus Wi-Fi?	It becomes very slow when many students are connected, especially in the evening.
3	Do you experience Wi-Fi dead zones?	Yes. The signal is weak in some hostel rooms and corridors.
4	Do video calls or YouTube videos have problems?	WhatsApp calls sometimes freeze, and YouTube videos buffer during busy periods.
5	What do you do when campus Wi-Fi is unavailable or slow?	I switch to mobile data, which increases my data usage and sometimes becomes expensive.

Top 5 Pain Points

1. Slow Internet During Peak Hours

When many students use the network simultaneously, internet speed decreases significantly.

2. Weak Wi-Fi Coverage

Some hostel rooms and corridors have weak signals or dead zones.

3. YouTube Buffering

Videos may pause or automatically reduce quality when network traffic is high.

4. WhatsApp Video Call Problems

Video calls can experience freezing, delays, or dropped connections.

5. High Mobile Data Usage

Students often switch to mobile data when campus Wi-Fi performs poorly, consuming their mobile data allowance.

Recommended Design Features

1. High-Density Wi-Fi Access Points

Install multiple **Wi-Fi 6 access points** throughout the hostel and campus instead of depending on a single router.

Benefits:

- Better coverage
- More users can connect simultaneously
- Less congestion per access point
- Better performance for YouTube and WhatsApp calls

2. Bandwidth Management and QoS

Configure the network with **Quality of Service (QoS)** and fair bandwidth allocation.

Benefits:

- Prevents a few users from consuming all available bandwidth
- Helps maintain stable video calls
- Reduces YouTube buffering during busy periods
- Provides fair access to all students

Additional useful features

- **Wireless site survey** to identify dead zones
- **Guest Wi-Fi** separated from the student network
- **Network monitoring** for detecting outages and overloaded APs
- **UPS backup** for networking equipment
- **Seamless roaming** between access points

Interview Conclusion

The interview showed that the main student concerns are **slow peak-hour speeds, poor coverage, video buffering, unstable video calls, and excessive mobile-data usage**. The proposed network should therefore prioritize **high-density Wi-Fi 6 access points, adequate internet capacity, QoS/bandwidth management, and good campus-wide coverage**.

Available next action: Create a downloadable PDF file here in this chat containing the findings and recommendations above.

5. Combine your requirements document and topology sketch into a single PDF file and submit it to your mentor or instructor for feedback.

Hint: Make sure your document is clear, organized, and uses simple diagrams or tables where possible.

-> Campus Wi-Fi Network Requirements & Topology

Selected daily-use app: Instagram

This document describes the requirements for designing a reliable campus Wi-Fi network for students and staff who regularly use Instagram and other internet-based applications. The design focuses on coverage, capacity, security, performance, and easy management.

1. Network Requirements

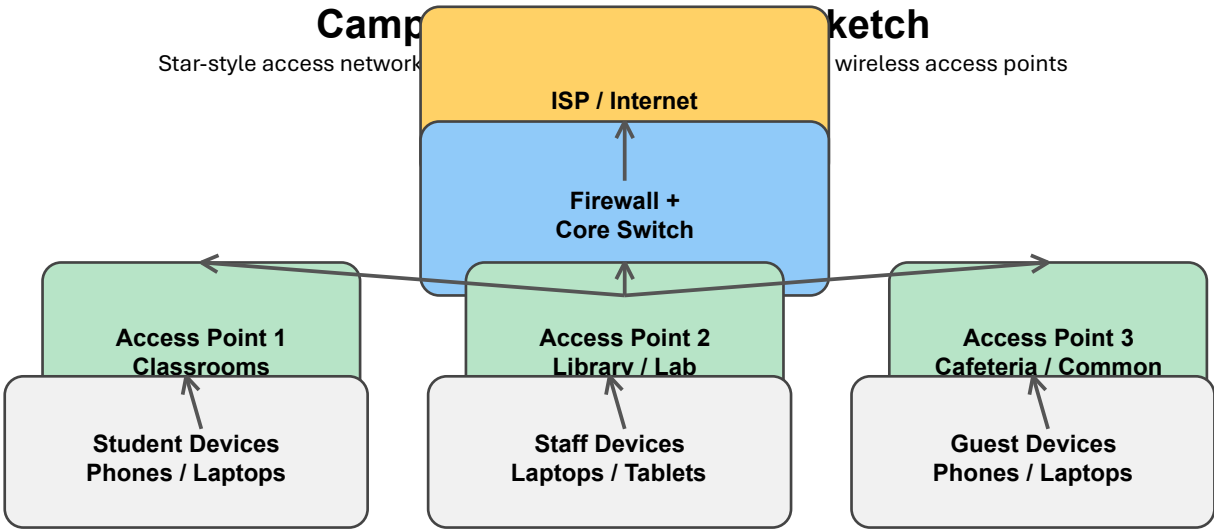
No.	Requirement	Description
1	Wide Wi-Fi coverage	Provide reliable coverage in classrooms, labs, library, corridors, cafeteria, and common areas.
2	High user capacity	Support many students and staff connecting at the same time without severe slowdowns.
3	High-speed internet	Provide sufficient bandwidth for Instagram photos/videos, web browsing, online classes, and cloud applications.
4	Strong authentication	Use WPA2/WPA3-Enterprise or another secure authentication method for authorized users.
5	Separate user networks	Use VLANs/SSIDs to separate students, staff, guests, and network-management traffic.
6	Guest Wi-Fi	Provide a separate guest network with internet access while preventing access to internal campus systems.
7	QoS and traffic management	Prioritize important academic traffic and manage high-bandwidth video/social-media usage fairly.
8	Reliable roaming	Allow mobile devices to move between access points with minimal interruption.
9	Monitoring and troubleshooting	Monitor access points, bandwidth, connected devices, and failures from a central management system.
10	Scalability	Allow additional access points, switches, and users to be added as the campus grows.

2. Proposed Network Components

Component	Purpose
ISP / Internet connection	Provides internet access to the campus.
Firewall / Gateway	Protects the network and controls internet traffic.

Core Layer-3 Switch	Connects major network segments and routes VLAN traffic.
PoE Access Switches	Connect wired network equipment and power compatible access points.
Wireless Access Points (APs)	Provide Wi-Fi coverage to users across campus areas.
Wi-Fi Controller / Cloud Management	Centralized AP configuration, monitoring, and troubleshooting.
Student / Staff / Guest devices	Laptops, smartphones, tablets, and other client devices.

3. Network Topology



Suggested logical separation:
Student VLAN • Staff VLAN • Guest VLAN • Network Management VLAN

Area	Plan
Authentication	Use strong WPA2/WPA3-Enterprise authentication for students and staff.
Guest access	Use a separate guest SSID/VLAN with internet-only access.
Segmentation	Separate student, staff, guest, and management traffic using VLANs.
Bandwidth	Use QoS, bandwidth limits, and monitoring to reduce congestion.

Monitoring	Check AP health, connected clients, bandwidth usage, and alerts regularly.
Maintenance	Keep firmware, switch software, and security settings updated.

Guest traffic is isolated from internal campus resources; all users receive controlled internet access.

How it works: The internet connection enters through the firewall/gateway and reaches the core switch. The core switch connects to multiple access points. Each AP serves a different campus area, while VLANs and security policies logically separate student, staff, guest, and management traffic.

4. Basic Security and Performance Plan

Conclusion: The proposed campus Wi-Fi design provides broad coverage, secure access, good capacity, traffic separation, and centralized management. It can be expanded by adding more access points and switching capacity as the number of users increases.